

Analyze Visium H&E data

Visium spots based purely on the tissue image

This tutorial shows how to apply Squidpy for the analysis of Visium spatial transcriptomics data.

https://squidpy.readthedocs.io/en/stable/notebooks/tutorials/tutorial_visium_hne.html

Data

dataset portal <https://support.10xgenomics.com/spatial-gene-expression/datasets>

```
In [1]: import numpy as np
import pandas as pd
import os, sys

import anndata as ad
import scanpy as sc
import squidpy as sq
import liana as li
from pathlib import Path

ROOT0 = Path("/home/flavio/uv/gwas/")
ROOT_SRC = ROOT0 / "src"

sys.path.append(str(ROOT_SRC))

from libs.Basic import create_dir, pdwritecsv, pdreadcsv
from libs.matplotlib_figure_style import *

import matplotlib.pyplot as plt

sc.logging.print_header()
print(f"squidpy=={sq.__version__}")

squidpy==1.8.4.dev25+g7a04f5884.d20260910
```

```
In [2]: # load the pre-processed dataset
img = sq.datasets.visium_hne_image()
```

```
In [3]: adata = sq.datasets.visium_hne_adata()
```

```
In [4]: root0 = Path('/home/flavio/uv/squidpy/')
root_data = root0 / 'data'
root_figure = root0 / 'figures'
root_result = root0 / 'results'

os.listdir(root_figure)
```

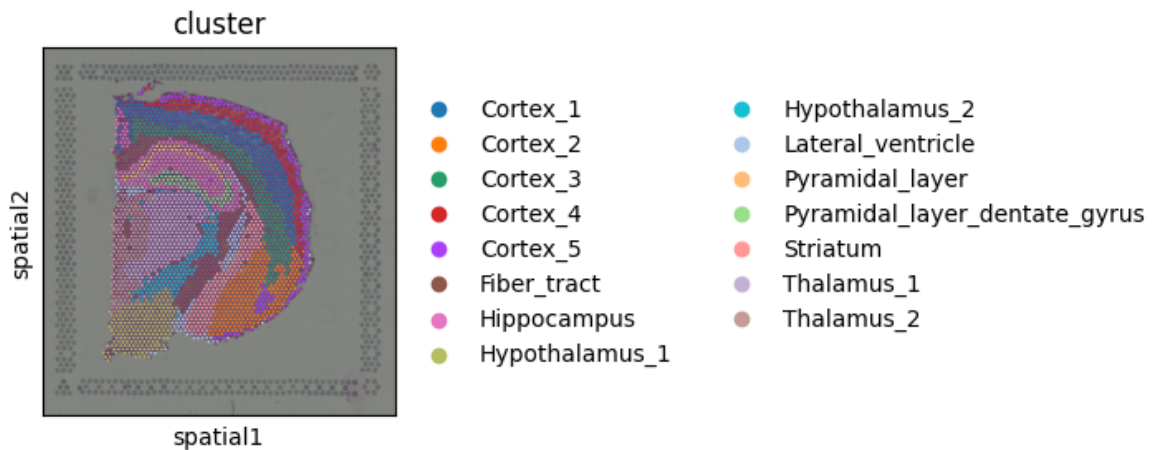
```
Out[4]: ['liana_cluster_dotplot.png', 'liana_spatial_LR.png']
```

```
In [5]: filename = root_data / "visium_hne.h5ad"
adata.write_h5ad(filename)
```

let's visualize cluster annotation in spatial context with `squidpy.pl.spatial_scatter()`.

```
In [6]: plt.rcParams['figure.figsize'] = (7,6)
# plt.rcParams['figure.dpi'] = 240

sq.pl.spatial_scatter(adata, color="cluster")
```



```
In [ ]:
```

Image feature

Visium datasets contain high-resolution images of the tissue that was used for the gene extraction. Using the function `squidpy.im.calculate_image_features()` you can calculate image features for each Visium spot and create a `obs x features` matrix in `adata` that can then be analyzed together with the `obs x gene` gene expression matrix.

By extracting image features we are aiming to get both similar and complementary information to the gene expression values. Similar information is for example present in the case of a tissue with two different cell types whose morphology is different. Such cell type information is then contained in both the gene expression values and the tissue image features.

Squidpy contains several feature extractors and a flexible pipeline of calculating features of different scales and sizes. There are several detailed examples of how to use `squidpy.im.calculate_image_features()`. Extract image features provides a good starting point for learning more.

```
In [ ]:
```

extract summary features at different crop sizes and scales

to allow the calculation of multi-scale features and segmentation features. For more information on the summary features, also refer to Extract [summary features](#).

```
In [7]: # calculate features for different scales (higher value means more context)
for scale in [1.0, 2.0]:
```

```

feature_name = f"features_summary_scale{scale}"
print(">>>", feature_name)
sq.im.calculate_image_features(
    adata,
    img.compute(),
    features="summary",
    key_added=feature_name,
    n_jobs=4,
    scale=scale,
)

# combine features in one dataframe
adata.obsm["features"] = pd.concat(
    [adata.obsm[f] for f in adata.obsm.keys() if "features_summary" in f]
    axis="columns",
)
# make sure that we have no duplicated feature names in the combined table
adata.obsm["features"].columns = ad.utils.make_index_unique(
    adata.obsm["features"].columns
)

>>> features_summary_scale1.0
0%|          | 0/2688 [00:00<?, ?/s]
>>> features_summary_scale2.0
0%|          | 0/2688 [00:00<?, ?/s]

```

In [8]:

```

for i, key in enumerate(adata.obsm.keys()):
    print(i, key)

```

```

0 X_pca
1 X_umap
2 spatial
3 features_summary_scale1.0
4 features_summary_scale2.0
5 features

```

extracted image features to compute a new cluster annotation

We can use the extracted image features to compute a new cluster annotation. This could be useful to gain insights in similarities across spots based on image morphology.

"features_cluster" are the "leinden" regions, what do they represent?

In that tutorial, `features_cluster` is a clustering of the **Visium spots based purely on the tissue image**, not on gene expression.

What's going on:

- `adata.obsm["features"]` comes from `sq.im.calculate_image_features(...)`. For each spot, Squidpy crops the H&E (or fluorescence) image around the spot's coordinates and computes numeric descriptors of that crop. The columns are prefixed by feature type: `summary_*`, `histogram_*`, `texture_*`, `segmentation_*`.

- `like="summary"` keeps only the summary-statistics features — per channel, things like mean, standard deviation, and quantiles of pixel intensity in the crop. So each spot is represented by a small vector describing how dark/light, how variable, and how colored its local patch of tissue looks.
- `cluster_features` then does the standard scanpy pipeline (scale → PCA → KNN graph → Leiden) on that matrix, treating spots as observations and image statistics as "genes".

So each Leiden label ("0", "1", "2", ...) is an arbitrary integer identifying a group of spots whose local histology looks alike in terms of staining intensity and contrast — e.g. densely packed, darkly stained nuclei-rich regions vs. pale, sparse white matter vs. background/edge spots. The labels carry no biological name and no spatial constraint; spots in the same cluster can be scattered across the section.

The point of the tutorial is the comparison with `adata.obs["cluster"]` (the expression-derived, anatomically annotated clusters): plotting both side by side shows how much of the anatomy is recoverable from image appearance alone. You typically see partial correspondence — large-scale structures like cortex vs. white matter vs. hippocampus separate reasonably well, while finer expression-defined subregions collapse or split differently. Adding `texture` or `segmentation` features (nuclei counts, cell density) usually improves that agreement.

```
In [9]: # helper function returning a clustering
def cluster_features(features: pd.DataFrame, like=None) -> pd.Series:
    """
    Calculate leiden clustering of features.
    Specify filter of features using `like`.
    """
    # filter features
    if like is not None:
        features = features.filter(like=like)
    # create temporary adata to calculate the clustering
    adata = ad.AnnData(features)
    # important - feature values are not scaled, so need to scale them before
    sc.pp.scale(adata)
    # calculate leiden clustering
    sc.pp.pca(adata, n_comps=min(10, features.shape[1] - 1))
    sc.pp.neighbors(adata)
    sc.tl.leiden(adata)

    return adata.obs["leiden"]
```

```
In [10]: # calculate feature clusters
adata.obs["features_cluster"] = cluster_features(adata.obs["features"],
```

/tmp/ipykernel_254612/3756014547.py:17: FutureWarning: The `igraph` implementation of leiden clustering is *orders of magnitude faster*. Set the flavor argument to (and install if needed) 'igraph' to use it.
In the future, the default backend for leiden will be igraph instead of leidenalg. To achieve the future defaults please pass: `flavor='igraph'` and `n\_iterations=2`. `directed` must also be `False` to work with igraph's implementation.

```
In [11]: for i, key in enumerate(adata.obsm.keys()):
         print(i, key)
```

```
0 X_pca
1 X_umap
2 spatial
3 features_summary_scale1.0
4 features_summary_scale2.0
5 features
```

```
In [12]: dfc = adata.obs['cluster']
         print(type(dfc), dfc.shape)
         dfc.head(6)
```

```
<class 'pandas.core.series.Series'> (2688,)
```

```
Out[12]: AAACAAGTATCTCCCA-1      Cortex_2
         AAACAATCTACTAGCA-1      Cortex_5
         AAACACCAATAACTGC-1      Thalamus_2
         AAACAGAGCGACTCCT-1      Cortex_5
         AAACCGGGTAGGTACC-1      Thalamus_2
         AAACCGTTCGTCCAGG-1      Hypothalamus_2
         Name: cluster, dtype: category
         Categories (15, object): ['Cortex_1', 'Cortex_2', 'Cortex_3', 'Cortex_4', ..., 'Pyramidal_layer_dentate_gyrus', 'Striatum', 'Thalamus_1', 'Thalamus_2']
```

There are 257 clusters for Cortex 2

```
In [13]: df2 = dfc[dfc=='Cortex_2']
         print(df2.shape)
         df2.head(3)
```

```
(257,)
```

```
Out[13]: AAACAAGTATCTCCCA-1      Cortex_2
         AAATAGGGTGCTATTG-1      Cortex_2
         AACACACGCTCGCCGC-1      Cortex_2
         Name: cluster, dtype: category
         Categories (15, object): ['Cortex_1', 'Cortex_2', 'Cortex_3', 'Cortex_4', ..., 'Pyramidal_layer_dentate_gyrus', 'Striatum', 'Thalamus_1', 'Thalamus_2']
```

```
In [14]: adata.obsm.__dict__
```

```

Out[14]: {'_axis': 0,
  '_parent': AnnData object with n_obs × n_vars = 2688 × 18078
    obs: 'in_tissue', 'array_row', 'array_col', 'n_genes_by_counts', 'loglp_n_genes_by_counts', 'total_counts', 'loglp_total_counts', 'pct_counts_in_top_50_genes', 'pct_counts_in_top_100_genes', 'pct_counts_in_top_200_genes', 'pct_counts_in_top_500_genes', 'total_counts_mt', 'loglp_total_counts_mt', 'pct_counts_mt', 'n_counts', 'leiden', 'cluster', 'features_cluster'
    var: 'gene_ids', 'feature_types', 'genome', 'mt', 'n_cells_by_counts', 'mean_counts', 'loglp_mean_counts', 'pct_dropout_by_counts', 'total_counts', 'loglp_total_counts', 'n_cells', 'highly_variable', 'highly_variable_rank', 'means', 'variances', 'variances_norm'
    uns: 'cluster_colors', 'hvg', 'leiden', 'leiden_colors', 'neighbors', 'pca', 'rank_genes_groups', 'spatial', 'umap'
    obsm: 'X_pca', 'X_umap', 'spatial', 'features_summary_scale1.0', 'features_summary_scale2.0', 'features'
    varm: 'PCs'
    obsp: 'connectivities', 'distances'
    layers: None (.X),
  '_data': {'X_pca': array([[ -9.6398516e+00, -2.9838152e+00,  2.0782962e-01, ...,
    -1.6275274e+00,  1.5929434e+00, -2.6020402e-01],
    [-1.2198642e+00, -1.9066552e+00, -2.9162614e+00, ...,
    -2.1797843e+00, -9.9834663e-01, -1.8425034e-03],
    [ 9.8286591e+00,  8.8669901e+00, -7.5049323e-01, ...,
    5.0805200e-02, -8.5250344e-03, -8.4113294e-01],
    ...,
    [ 4.9666276e+00,  1.3346644e+01, -1.8030597e+00, ...,
    3.0393279e-01,  4.5136851e-01, -1.1242243e+00],
    [ 5.7296743e+00,  2.0515683e+00,  1.0510375e+01, ...,
    -4.5647275e-01,  3.0606148e-01,  5.4324287e-01],
    [-1.2606829e+01, -3.4644568e+00, -2.3527545e-01, ...,
    3.5502258e-01, -6.7781830e-01,  1.0772370e+00]],
    shape=(2688, 50), dtype=float32),
  'X_umap': array([[ -0.48594925, -1.3089514 ],
    [-0.6433954 ,  1.128548 ],
    [11.860388 ,  1.2172521 ],
    ...,
    [10.546704 ,  0.42895854],
    [16.20739 ,  1.0817059 ],
    [-4.0604615 , -0.46168917]], shape=(2688, 2), dtype=float32),
  'spatial': array([[8230, 7237],
    [4170, 1611],
    [2519, 8315],
    ...,
    [3276, 8435],
    [3069, 6639],
    [4720, 2090]]), shape=(2688, 2)),
  'features_summary_scale1.0':
-0.9 summary_ch-0_quantile-0.5 \
AAACAAGTATCTCCCA-1 132.0 11
1.0
AAACAATCTACTAGCA-1 140.0 11
1.0
AAACACCAATAACTGC-1 132.0 11
6.0
AAACAGAGCGACTCCT-1 136.0 11
6.0
AAACCGGGTAGGTACC-1 137.0 11
2.0

```

...	...	
...		
TTGTTGTGTGTCAAGA-1	135.0	10
7.0		
TTGTTTCACATCCAGG-1	139.0	10
7.0		
TTGTTTCATTAGTCTA-1	135.0	11
1.0		
TTGTTTCCATACAAC-1	136.0	11
0.0		
TTGTTTGTGTAAATTC-1	139.0	11
5.0		

	summary_ch-0_quantile-0.1	summary_ch-0_mean	\
AAACAAGTATCTCCCA-1	77.0	107.571140	
AAACAATCTACTAGCA-1	80.0	109.815175	
AAACACCAATAACTGC-1	89.0	112.421285	
AAACAGAGCGACTCCT-1	93.0	115.085217	
AAACCGGGTAGGTACC-1	83.0	110.426840	
...	...	...	
TTGTTGTGTGTCAAGA-1	45.0	98.505365	
TTGTTTCACATCCAGG-1	64.0	104.259942	
TTGTTTCATTAGTCTA-1	73.0	107.579220	
TTGTTTCCATACAAC-1	77.0	107.720490	
TTGTTTGTGTAAATTC-1	81.0	111.916930	

	summary_ch-0_std	summary_ch-1_quantile-0.9	\
AAACAAGTATCTCCCA-1	21.767668	102.0	
AAACAATCTACTAGCA-1	25.047208	87.0	
AAACACCAATAACTGC-1	19.088559	109.0	
AAACAGAGCGACTCCT-1	17.450778	115.0	
AAACCGGGTAGGTACC-1	21.210674	103.0	
...	...	...	
TTGTTGTGTGTCAAGA-1	33.255413	104.0	
TTGTTTCACATCCAGG-1	28.099418	112.0	
TTGTTTCATTAGTCTA-1	24.471029	107.0	
TTGTTTCCATACAAC-1	23.502833	99.0	
TTGTTTGTGTAAATTC-1	24.157025	122.0	

	summary_ch-1_quantile-0.5	summary_ch-1_quantile-
0.1 \		
AAACAAGTATCTCCCA-1	79.0	5
4.0		
AAACAATCTACTAGCA-1	60.0	3
9.0		
AAACACCAATAACTGC-1	91.0	6
7.0		
AAACAGAGCGACTCCT-1	82.0	5
8.0		
AAACCGGGTAGGTACC-1	78.0	5
4.0		
...	...	
...		
TTGTTGTGTGTCAAGA-1	74.0	3
3.0		
TTGTTTCACATCCAGG-1	81.0	4
9.0		
TTGTTTCATTAGTCTA-1	81.0	5
4.0		
TTGTTTCCATACAAC-1	73.0	4

9.0			
	TTGTTTGTGTAAATTC-1	93.0	6
2.0			
		summary_ch-1_mean	summary_ch-1_std \
	AAACAAGTATCTCCCA-1	78.586668	19.153915
	AAACAATCTACTAGCA-1	61.846105	19.495044
	AAACACCAATAACTGC-1	89.382780	16.884106
	AAACAGAGCGACTCCT-1	84.919202	22.004344
	AAACCGGGTAGGTACC-1	78.434920	19.046626
	...	...	...
	TTGTTGTGTGTCAAGA-1	71.122964	26.024651
	TTGTTTCACATCCAGG-1	80.879687	24.025899
	TTGTTTCATTAGTCTA-1	80.455372	20.784933
	TTGTTTCCATACAAC-1	73.409923	19.286326
	TTGTTTGTGTAAATTC-1	92.590077	24.090925
		summary_ch-2_quantile-0.9	summary_ch-2_quantile-
0.5 \			
	AAACAAGTATCTCCCA-1	138.0	11
1.0			
	AAACAATCTACTAGCA-1	135.0	10
8.0			
	AAACACCAATAACTGC-1	130.0	11
7.0			
	AAACAGAGCGACTCCT-1	137.0	11
3.0			
	AAACCGGGTAGGTACC-1	133.0	11
3.0			
	...	...	
...			
	TTGTTGTGTGTCAAGA-1	143.0	11
1.0			
	TTGTTTCACATCCAGG-1	140.0	11
3.0			
	TTGTTTCATTAGTCTA-1	134.0	11
4.0			
	TTGTTTCCATACAAC-1	133.0	11
2.0			
	TTGTTTGTGTAAATTC-1	140.0	11
8.0			
		summary_ch-2_quantile-0.1	summary_ch-2_mean \
	AAACAAGTATCTCCCA-1	88.0	112.461305
	AAACAATCTACTAGCA-1	84.0	108.806211
	AAACACCAATAACTGC-1	97.0	114.699912
	AAACAGAGCGACTCCT-1	89.0	113.176998
	AAACCGGGTAGGTACC-1	92.0	112.872996
	...	...	...
	TTGTTGTGTGTCAAGA-1	74.0	109.601818
	TTGTTTCACATCCAGG-1	84.0	112.594622
	TTGTTTCATTAGTCTA-1	89.0	112.461305
	TTGTTTCCATACAAC-1	88.0	110.888272
	TTGTTTGTGTAAATTC-1	92.0	117.067794
		summary_ch-2_std	
	AAACAAGTATCTCCCA-1	19.733864	
	AAACAATCTACTAGCA-1	19.869577	
	AAACACCAATAACTGC-1	13.826571	
	AAACAGAGCGACTCCT-1	18.714021	

```

AAACCGGGTAGGTACC-1      15.922622
...
TTGTTGTGTGTCAAGA-1      26.373283
TTGTTTCACATCCAGG-1      21.763193
TTGTTTCATTAGTCTA-1      18.158389
TTGTTTCCATACAAC-1       18.042013
TTGTTTGTGTAAATTC-1      19.352681

[2688 rows x 15 columns],
'features_summary_scale2.0':
-0.9 summary_ch-0_quantile-0.5 \
AAACAAGTATCTCCCA-1      129.0      11
1.0
AAACAATCTACTAGCA-1      138.0      11
0.0
AAACACCAATAACTGC-1      131.0      11
5.0
AAACAGAGCGACTCCT-1      132.7      11
6.0
AAACCGGGTAGGTACC-1      135.0      11
1.0
...
...
TTGTTGTGTGTCAAGA-1      132.0      10
7.0
TTGTTTCACATCCAGG-1      137.0      10
6.0
TTGTTTCATTAGTCTA-1      134.0      11
1.0
TTGTTTCCATACAAC-1       134.0      11
0.0
TTGTTTGTGTAAATTC-1      137.0      11
4.0

summary_ch-0_quantile-0.1 summary_ch-0_mean \
AAACAAGTATCTCCCA-1      78.0      107.103585
AAACAATCTACTAGCA-1      81.0      109.338341
AAACACCAATAACTGC-1      90.0      111.950574
AAACAGAGCGACTCCT-1      96.0      114.627572
AAACCGGGTAGGTACC-1      84.0      109.960548
...
...
TTGTTGTGTGTCAAGA-1      45.0      98.033676
TTGTTTCACATCCAGG-1      65.0      103.783645
TTGTTTCATTAGTCTA-1      74.0      107.112233
TTGTTTCCATACAAC-1       78.0      107.247917
TTGTTTGTGTAAATTC-1      82.0      111.445051

summary_ch-0_std summary_ch-1_quantile-0.9 \
AAACAAGTATCTCCCA-1      20.010766      99.0
AAACAATCTACTAGCA-1      23.999486      85.0
AAACACCAATAACTGC-1      18.595623      108.0
AAACAGAGCGACTCCT-1      15.426818      114.0
AAACCGGGTAGGTACC-1      20.283293      102.0
...
...
TTGTTGTGTGTCAAGA-1      32.056541      101.0
TTGTTTCACATCCAGG-1      27.083072      110.0
TTGTTTCATTAGTCTA-1      23.654244      105.0
TTGTTTCCATACAAC-1       22.613715      97.0
TTGTTTGTGTAAATTC-1      22.954867      120.0

```

	summary_ch-1_quantile-0.5	summary_ch-1_quantile-
0.1 \		
AAACAAGTATCTCCCA-1	79.0	5
6.0		
AAACAATCTACTAGCA-1	60.0	3
9.0		
AAACACCAATAACTGC-1	90.0	6
8.0		
AAACAGAGCGACTCCT-1	82.0	6
0.0		
AAACCGGGTAGGTACC-1	78.0	5
5.0		
...	...	
...		
TTGTTGTGTGTCAAGA-1	74.0	3
4.0		
TTGTTTCACATCCAGG-1	80.0	5
0.0		
TTGTTTCATTAGTCTA-1	81.0	5
4.0		
TTGTTTCCATACAAC-1	73.0	4
9.0		
TTGTTTGTGTAAATTC-1	93.0	6
3.0		

	summary_ch-1_mean	summary_ch-1_std \
AAACAAGTATCTCCCA-1	78.123248	17.307853
AAACAATCTACTAGCA-1	61.372806	18.517388
AAACACCAATAACTGC-1	88.907493	16.411915
AAACAGAGCGACTCCT-1	84.458654	20.596870
AAACCGGGTAGGTACC-1	77.971342	18.140652
...	...	...
TTGTTGTGTGTCAAGA-1	70.653421	24.716731
TTGTTTCACATCCAGG-1	80.408945	22.981237
TTGTTTCATTAGTCTA-1	79.989774	19.979603
TTGTTTCCATACAAC-1	72.940759	18.380322
TTGTTTGTGTAAATTC-1	92.120723	22.966558

	summary_ch-2_quantile-0.9	summary_ch-2_quantile-
0.5 \		
AAACAAGTATCTCCCA-1	134.0	11
2.0		
AAACAATCTACTAGCA-1	133.0	10
8.0		
AAACACCAATAACTGC-1	129.0	11
6.0		
AAACAGAGCGACTCCT-1	133.0	11
3.0		
AAACCGGGTAGGTACC-1	132.0	11
3.0		
...	...	
...		
TTGTTGTGTGTCAAGA-1	140.0	11
1.0		
TTGTTTCACATCCAGG-1	138.0	11
3.0		
TTGTTTCATTAGTCTA-1	133.0	11
3.0		
TTGTTTCCATACAAC-1	132.0	11
1.0		

TTGTTTGTGTAAATTC-1	138.0	11
--------------------	-------	----

8.0

	summary_ch-2_quantile-0.1	summary_ch-2_mean	\
AAACAAGTATCTCCCA-1	90.0	111.994919	
AAACAATCTACTAGCA-1	85.0	108.325969	
AAACACCAATAACTGC-1	98.0	114.225193	
AAACAGAGCGACTCCT-1	91.0	112.717776	
AAACCGGGTAGGTACC-1	93.0	112.406641	
...	...	...	
TTGTTGTGTGTCAAGA-1	75.0	109.125710	
TTGTTTCACATCCAGG-1	85.0	112.121576	
TTGTTTCATTAGTCTA-1	89.0	111.993183	
TTGTTTCCATACAAC-1	89.0	110.416835	
TTGTTTGTGTAAATTC-1	93.0	116.599703	

	summary_ch-2_std
AAACAAGTATCTCCCA-1	17.550044
AAACAATCTACTAGCA-1	18.951807
AAACACCAATAACTGC-1	13.425479
AAACAGAGCGACTCCT-1	16.685592
AAACCGGGTAGGTACC-1	15.151956
...	...
TTGTTGTGTGTCAAGA-1	24.864237
TTGTTTCACATCCAGG-1	20.841642
TTGTTTCATTAGTCTA-1	17.442640
TTGTTTCCATACAAC-1	17.267120
TTGTTTGTGTAAATTC-1	18.169677

[2688 rows x 15 columns],
'features':

	summary_ch-0_quantile-0.9	summary_ch-
0_quantile-0.5 \		
AAACAAGTATCTCCCA-1	132.0	11
1.0		
AAACAATCTACTAGCA-1	140.0	11
1.0		
AAACACCAATAACTGC-1	132.0	11
6.0		
AAACAGAGCGACTCCT-1	136.0	11
6.0		
AAACCGGGTAGGTACC-1	137.0	11
2.0		
...	...	
...		
TTGTTGTGTGTCAAGA-1	135.0	10
7.0		
TTGTTTCACATCCAGG-1	139.0	10
7.0		
TTGTTTCATTAGTCTA-1	135.0	11
1.0		
TTGTTTCCATACAAC-1	136.0	11
0.0		
TTGTTTGTGTAAATTC-1	139.0	11
5.0		

	summary_ch-0_quantile-0.1	summary_ch-0_mean	\
AAACAAGTATCTCCCA-1	77.0	107.571140	
AAACAATCTACTAGCA-1	80.0	109.815175	
AAACACCAATAACTGC-1	89.0	112.421285	
AAACAGAGCGACTCCT-1	93.0	115.085217	

AAACCGGGTAGGTACC-1	83.0	110.426840	
...	...	...	
TTGTTGTGTGTCAAGA-1	45.0	98.505365	
TTGTTTCACATCCAGG-1	64.0	104.259942	
TTGTTTCATTAGTCTA-1	73.0	107.579220	
TTGTTTCCATACAAC-1	77.0	107.720490	
TTGTTTGTGTAAATTC-1	81.0	111.916930	
	summary_ch-0_std	summary_ch-1_quantile-0.9	\
AAACAAGTATCTCCCA-1	21.767668	102.0	
AAACAATCTACTAGCA-1	25.047208	87.0	
AAACACCAATAACTGC-1	19.088559	109.0	
AAACAGAGCGACTCCT-1	17.450778	115.0	
AAACCGGGTAGGTACC-1	21.210674	103.0	
...	...	...	
TTGTTGTGTGTCAAGA-1	33.255413	104.0	
TTGTTTCACATCCAGG-1	28.099418	112.0	
TTGTTTCATTAGTCTA-1	24.471029	107.0	
TTGTTTCCATACAAC-1	23.502833	99.0	
TTGTTTGTGTAAATTC-1	24.157025	122.0	
	summary_ch-1_quantile-0.5	summary_ch-1_quantile-	
0.1 \			
AAACAAGTATCTCCCA-1	79.0	5	
4.0			
AAACAATCTACTAGCA-1	60.0	3	
9.0			
AAACACCAATAACTGC-1	91.0	6	
7.0			
AAACAGAGCGACTCCT-1	82.0	5	
8.0			
AAACCGGGTAGGTACC-1	78.0	5	
4.0			
...	...		
...			
TTGTTGTGTGTCAAGA-1	74.0	3	
3.0			
TTGTTTCACATCCAGG-1	81.0	4	
9.0			
TTGTTTCATTAGTCTA-1	81.0	5	
4.0			
TTGTTTCCATACAAC-1	73.0	4	
9.0			
TTGTTTGTGTAAATTC-1	93.0	6	
2.0			
	summary_ch-1_mean	summary_ch-1_std	... \
AAACAAGTATCTCCCA-1	78.586668	19.153915	...
AAACAATCTACTAGCA-1	61.846105	19.495044	...
AAACACCAATAACTGC-1	89.382780	16.884106	...
AAACAGAGCGACTCCT-1	84.919202	22.004344	...
AAACCGGGTAGGTACC-1	78.434920	19.046626	...
...	...	...	...
TTGTTGTGTGTCAAGA-1	71.122964	26.024651	...
TTGTTTCACATCCAGG-1	80.879687	24.025899	...
TTGTTTCATTAGTCTA-1	80.455372	20.784933	...
TTGTTTCCATACAAC-1	73.409923	19.286326	...
TTGTTTGTGTAAATTC-1	92.590077	24.090925	...
	summary_ch-1_quantile-0.9-1	summary_ch-1_quantile	

-0.5-1 \	
AAACAAGTATCTCCCA-1	99.0
79.0	
AAACAATCTACTAGCA-1	85.0
60.0	
AAACACCAATAACTGC-1	108.0
90.0	
AAACAGAGCGACTCCT-1	114.0
82.0	
AAACCGGGTAGGTACC-1	102.0
78.0	
...	...
...	
TTGTTGTGTGTCAAGA-1	101.0
74.0	
TTGTTTCACATCCAGG-1	110.0
80.0	
TTGTTTCATTAGTCTA-1	105.0
81.0	
TTGTTTCCATACAAC-1	97.0
73.0	
TTGTTTGTGTAAATTC-1	120.0
93.0	

	summary_ch-1_quantile-0.1-1	summary_ch-1_mean-1
\		
AAACAAGTATCTCCCA-1	56.0	78.123248
AAACAATCTACTAGCA-1	39.0	61.372806
AAACACCAATAACTGC-1	68.0	88.907493
AAACAGAGCGACTCCT-1	60.0	84.458654
AAACCGGGTAGGTACC-1	55.0	77.971342
...	...	...
TTGTTGTGTGTCAAGA-1	34.0	70.653421
TTGTTTCACATCCAGG-1	50.0	80.408945
TTGTTTCATTAGTCTA-1	54.0	79.989774
TTGTTTCCATACAAC-1	49.0	72.940759
TTGTTTGTGTAAATTC-1	63.0	92.120723

	summary_ch-1_std-1	summary_ch-2_quantile-0.9-1	\
AAACAAGTATCTCCCA-1	17.307853	134.0	
AAACAATCTACTAGCA-1	18.517388	133.0	
AAACACCAATAACTGC-1	16.411915	129.0	
AAACAGAGCGACTCCT-1	20.596870	133.0	
AAACCGGGTAGGTACC-1	18.140652	132.0	
...	...	...	
TTGTTGTGTGTCAAGA-1	24.716731	140.0	
TTGTTTCACATCCAGG-1	22.981237	138.0	
TTGTTTCATTAGTCTA-1	19.979603	133.0	
TTGTTTCCATACAAC-1	18.380322	132.0	
TTGTTTGTGTAAATTC-1	22.966558	138.0	

	summary_ch-2_quantile-0.5-1	summary_ch-2_quantile
-0.1-1 \		
AAACAAGTATCTCCCA-1	112.0	
90.0		
AAACAATCTACTAGCA-1	108.0	
85.0		
AAACACCAATAACTGC-1	116.0	
98.0		
AAACAGAGCGACTCCT-1	113.0	

```

91.0
  AAACCGGGTAGGTACC-1      113.0
93.0
  ...
  ...
  TTGTTGTGTGTCAAGA-1      111.0
75.0
  TTGTTTCACATCCAGG-1      113.0
85.0
  TTGTTTCATTAGTCTA-1      113.0
89.0
  TTGTTTCCATACAAC-1       111.0
89.0
  TTGTTTGTGTAAATTC-1      118.0
93.0

```

	summary_ch-2_mean-1	summary_ch-2_std-1
AAACAAGTATCTCCCA-1	111.994919	17.550044
AAACAATCTACTAGCA-1	108.325969	18.951807
AAACACCAATAACTGC-1	114.225193	13.425479
AAACAGAGCGACTCCT-1	112.717776	16.685592
AAACCGGGTAGGTACC-1	112.406641	15.151956
...	...	...
TTGTTGTGTGTCAAGA-1	109.125710	24.864237
TTGTTTCACATCCAGG-1	112.121576	20.841642
TTGTTTCATTAGTCTA-1	111.993183	17.442640
TTGTTTCCATACAAC-1	110.416835	17.267120
TTGTTTGTGTAAATTC-1	116.599703	18.169677

```
[2688 rows x 30 columns]]}
```

```
In [15]: df = adata.X
         print(df.shape)
         df
```

```
(2688, 18078)
```

```
Out[15]: <Compressed Sparse Row sparse matrix of dtype 'float32'
         with 15411927 stored elements and shape (2688, 18078)>
```

```
In [16]: # obs: 'in_tissue', 'array_row', 'array_col', 'n_genes_by_counts', 'loglp
# var: 'gene_ids', 'feature_types', 'genome', 'mt', 'n_cells_by_counts',

fig, axes = plt.subplots(2, 2, figsize=(12, 4), sharex=True, sharey=True)

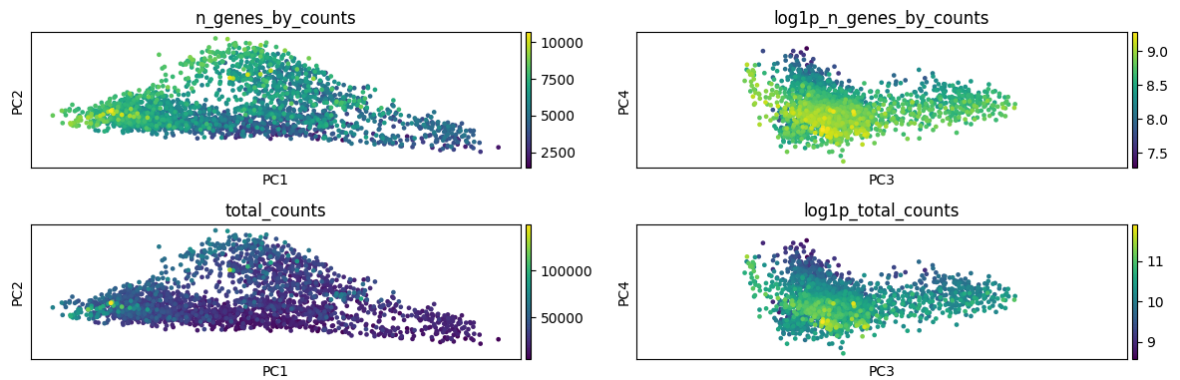
for ax, color, dimensions in zip(axes.flat, ["n_genes_by_counts", "loglp_
sc.pl.pca(adata, color=color, dimensions=dimensions, ax=ax, show=False)
print(f"Plotting PCA for '{color}' with dimensions {dimensions}")

plt.tight_layout()
```

```

Plotting PCA for 'n_genes_by_counts' with dimensions (0, 1)
Plotting PCA for 'loglp_n_genes_by_counts' with dimensions (2, 3)
Plotting PCA for 'total_counts' with dimensions (0, 1)
Plotting PCA for 'loglp_total_counts' with dimensions (2, 3)

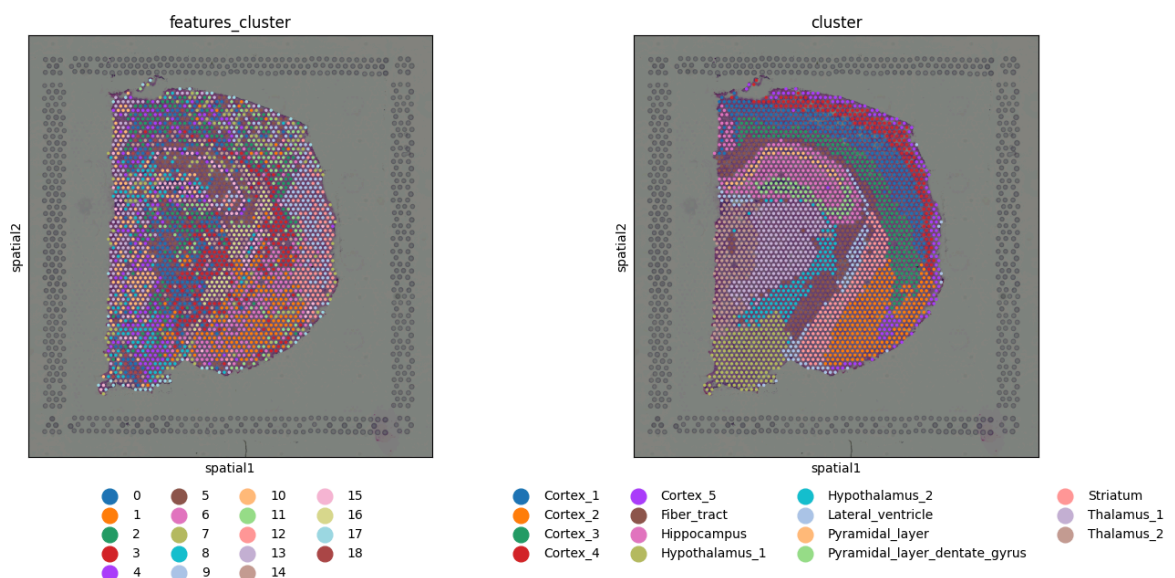
```



```
In [17]: # compare feature and gene clusters
# sq.pl.spatial_scatter(adata, color=["features_cluster", "cluster"])

axs = sq.pl.spatial_scatter(
    adata,
    color=["features_cluster", "cluster"],
    return_ax=True,
)
axs = [axs] if not isinstance(axs, (list, tuple)) else axs

for ax in axs:
    leg = ax.get_legend()
    if leg is None:
        continue
    handles = leg.legend_handles # .legendHandles on matplotlib
    labels = [t.get_text() for t in leg.get_texts()]
    leg.remove()
    ax.legend(
        handles, labels,
        loc="upper center", bbox_to_anchor=(0.5, -0.05),
        ncol=4, frameon=False, markerscale=2,
    )
```



Comparing gene and feature clusters, we notice that in some regions, they look very similar, like the cluster Fiber_tract, or clusters around the Hippocampus seems to be roughly recapitulated by the clusters in image feature space. In others, the feature clusters look different, like in the cortex, where the gene clusters show the layered

structure of the cortex, and the features clusters rather seem to show different regions of the cortex.

This is only a simple, comparative analysis of the image features, note that you could also use the image features to e.g. compute a common image and gene clustering by computing a shared neighbors graph (for instance on concatenated PCAs on both feature spaces).

In []:

```
In [18]: arr_spatial = adata.obsm['spatial']
arr_spatial
```

```
Out[18]: array([[8230, 7237],
                [4170, 1611],
                [2519, 8315],
                ...,
                [3276, 8435],
                [3069, 6639],
                [4720, 2090]], shape=(2688, 2))
```

```
In [19]: dff = adata.obsm['features']
print(dff.shape)
dff.iloc[:3, :6]
```

(2688, 30)

```
Out[19]:
```

	summary_ch- 0_quantile- 0.9	summary_ch- 0_quantile- 0.5	summary_ch- 0_quantile- 0.1	summary_ch- 0_mean	summary_ch- 0_std
AAACAAGTATCTCCCA- 1	132.0	111.0	77.0	107.571140	21.7
AAACAATCTACTAGCA- 1	140.0	111.0	80.0	109.815175	25.0
AAACACCAATAACTGC- 1	132.0	116.0	89.0	112.421285	19.0



In []:

Save complete h5ad

```
In [20]: filename = root_data / "visium_hne_complete.h5ad"
adata.write_h5ad(filename)
```

In []:

Liana (Saez lab - Heidelberg)

<https://github.com/saezlab/liana>

A. Cluster-level (non-spatial) LR inference

```
In [21]: li.mt.rank_aggregate(
    adata,
    groupby="cluster",          # anatomical spot clusters (15 regio
    resource_name="mouseconsensus", # mouse gene symbols
    expr_prop=0.1,
    use_raw=False,              # .raw holds counts here!
    n_perms=1000, n_jobs=8,
    verbose=True,
)
```

Using resource `mouseconsensus`.

Using `.X`!

/home/flavio/uv/squidpy/.venv/lib/python3.12/site-packages/liana/method/_pipe_utils/_pre.py:167: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

0.18 of entities in the resource are missing from the data.

Generating ligand-receptor stats for 2688 samples and 1257 features

/home/flavio/.local/share/uv/python/cpython-3.12.12-linux-x86_64-gnu/lib/python3.12/functools.py:912: UserWarning: zero-centering a sparse array/matrix densifies it.

/home/flavio/uv/squidpy/.venv/lib/python3.12/site-packages/liana/method/sc/_liana_pipe.py:288: ImplicitModificationWarning: Setting element `.layers['scaled']` of view, initializing view as actual.

/home/flavio/uv/squidpy/.venv/lib/python3.12/site-packages/liana/method/sc/_liana_pipe.py:293: FutureWarning: The method uns_keys is deprecated and will be removed in the future. Use uns instead of uns_keys. (e.g. `k in adata.uns` or `sorted(adata.uns)`)

/home/flavio/uv/squidpy/.venv/lib/python3.12/site-packages/liana/method/sc/_liana_pipe.py:296: FutureWarning: The method uns_keys is deprecated and will be removed in the future. Use uns instead of uns_keys. (e.g. `k in adata.uns` or `sorted(adata.uns)`)

Assuming that counts were `natural` log-normalized!

Running CellPhoneDB

100%|██████████| 1000/1000 [00:05<00:00, 187.65it/s]

Running Connectome

Running log2FC

Running NATMI

Running SingleCellSignalR

**source, target, ligand_complex, receptor_complex,
...magnitude_rank, specificity_rank (lower = stronger)**

```
In [22]: res = adata.uns["liana_res"]
pdwritecsv(res, "liana_cluster_results.tsv", root_result, verbose=True)
```

Table saved ((190666, 13)) at '/home/flavio/uv/squidpy/results/liana_cluster_results.tsv'

Out[22]: True

```
In [23]: top_specific = res[(res.specificity_rank <= 0.01)].sort_values("magnitude")
pdwritecsv(top_specific, "liana_cluster_top_specific.tsv", root_result, v

print(top_specific.shape)
top_specific.head(6)
```

Table saved ((25341, 13)) at '/home/flavio/uv/squidpy/results/liana_cluster_top_specific.tsv'
(25341, 13)

Out[23]:

	source	target	ligand_complex	receptor_complex	lr_means	cellp
130723	Pyramidal_layer	Fiber_tract	App	Aplp1	3.613924	
65802	Fiber_tract	Fiber_tract	App	Aplp1	3.563859	
4024	Cortex_1	Fiber_tract	App	Aplp1	3.563744	
29636	Cortex_3	Fiber_tract	App	Aplp1	3.553724	
170893	Thalamus_1	Fiber_tract	App	Aplp1	3.532249	
42170	Cortex_4	Fiber_tract	App	Aplp1	3.531248	

In []:

B. Spatially-resolved LR (per spot, no clusters needed)

- Spot centre spacing here is ~137 px (~100 um); bandwidth=200 px ~ 1-2 rings of neighbours.

```
In [25]: li.ut.spatial_neighbors(adata, bandwidth=200, cutoff=0.1, kernel="gaussia

lrdata = li.mt.bivariate(
    adata, resource_name="mouseconsensus",
    local_name="cosine",      # local (per-spot) LR co-localisation score
    global_name="morans",    # global spatial autocorrelation per LR -> l
    n_perms=100, mask_negatives=False, add_categories=True,
    nz_prop=0.2, use_raw=False, verbose=True,
)
# lrdata: spots x LR-pairs AnnData; .var has morans / morans_pvals; .laye
top_global = lrdata.var.sort_values("morans", ascending=False)
```

Using `X`!

Using resource `mouseconsensus`.

100%|██████████| 100/100 [00:01<00:00, 75.77it/s]

100%|██████████| 100/100 [00:04<00:00, 20.28it/s]

```
In [27]: print(top_global.shape)
top_global.head(6)
```

(454, 10)

Out[27]:

	ligand	receptor	ligand_means	ligand_props	receptor_means	recep
interaction						
Nptx1^Nptxr	Nptx1	Nptxr	1.045611	0.723586	1.783589	
Rims1^Slc17a7	Rims1	Slc17a7	0.628196	0.638393	2.257496	
Cck^Cckbr	Cck	Cckbr	2.622576	0.947917	0.295722	
Sema4d^Plxnb3	Sema4d	Plxnb3	0.544131	0.524554	0.190867	
Rtn4^Lingo1	Rtn4	Lingo1	2.446542	0.992188	1.553612	
Rtn4^Rtn4r	Rtn4	Rtn4r	2.446542	0.992188	0.675537	

spatial (bivariate) results: the ligand^receptor name is the INDEX, so keep it

```
In [28]: pdwritecsv(lrdata.var.sort_values("morans", ascending=False),
               "liana_spatial_global_morans.tsv", root_result, index=True, ve
```

Table saved ((454, 10)) at '/home/flavio/uv/squidpy/results/liana_spatial_global_morans.tsv'

Out[28]: True

Reading a file back

Tested sizes: the full table is 190,666 rows. The specificity_rank ≤ 0.01 filter keeps 25,341 rows, and the Moran's I table has 454 rows.

```
In [29]: dfqq = pdreadcsv("liana_cluster_top_specific.tsv", root_result)
dfqq.shape
```

Out[29]: (25341, 13)

Dotplot

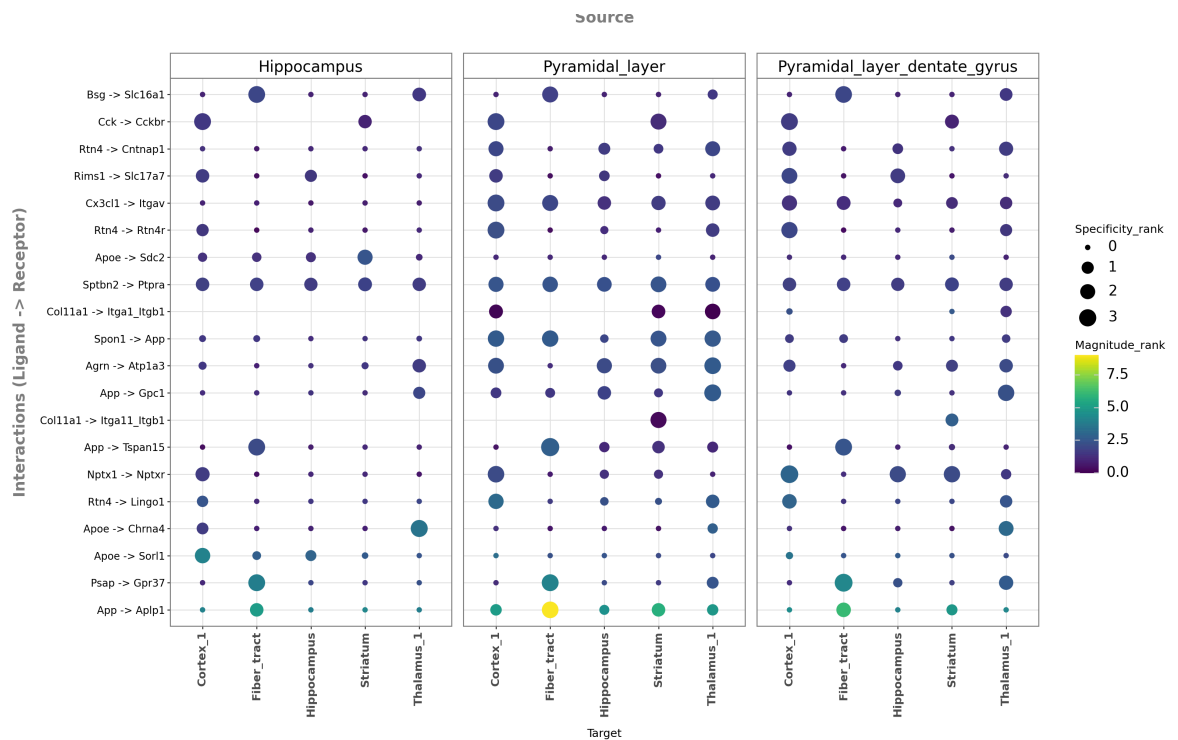
```
In [32]: p = li.pl.dotplot(adata=adata, colour="magnitude_rank", size="specificity
               inverse_colour=True, inverse_size=True,
               source_labels=["Hippocampus", "Pyramidal_layer", "Pyram
               target_labels=["Cortex_1", "Thalamus_1", "Fiber_tract",
               filter_fun=lambda x: x["specificity_rank"] <= 0.01,
               top_n=20, orderby="magnitude_rank", orderby_ascending=T

p.save(root_figure / "liana_cluster_dotplot.png", dpi=300, verbose=False)
```

/home/flavio/uv/squidpy/.venv/lib/python3.12/site-packages/liana/plotting/_common.py:98: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

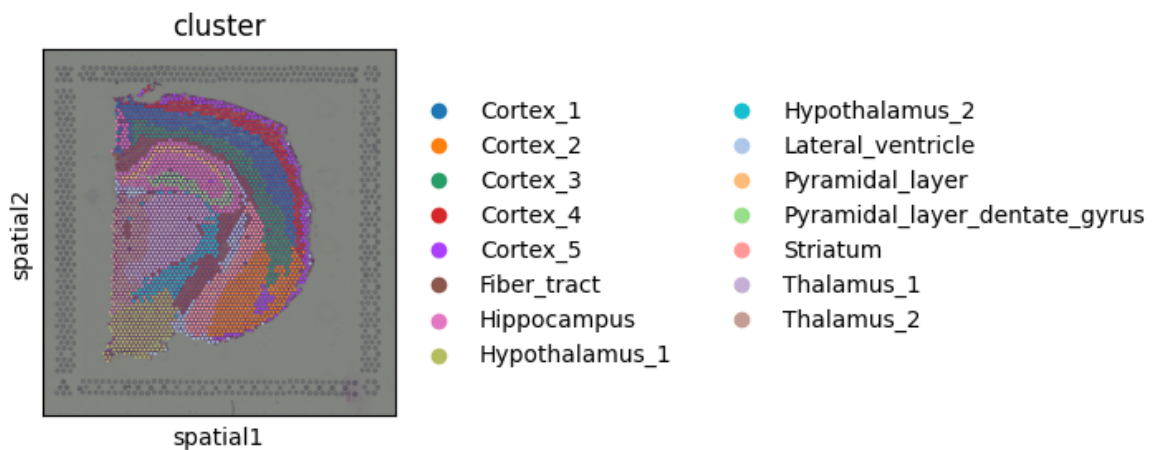
```
In [33]: p.show()
```



Squidpy and Scanpy plots.

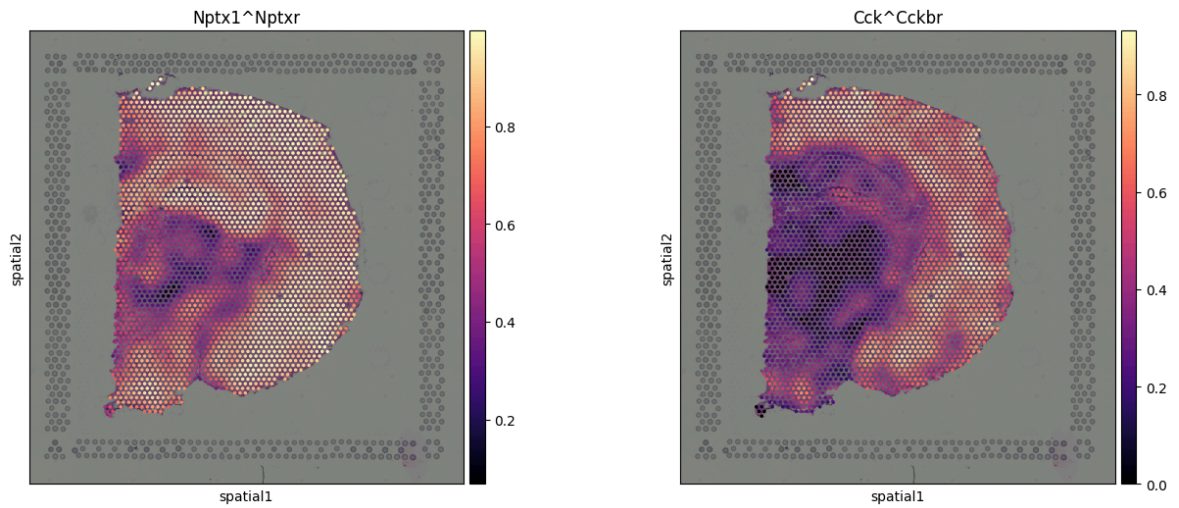
Get the axes back with `return_ax=True`, then save the parent figure:

```
In [34]: ax = sq.pl.spatial_scatter(adata, color="cluster", return_ax=True)
ax.figure.savefig(root_figure / "spatial_clusters.png", dpi=300, bbox_inches='tight')
plt.show()
```

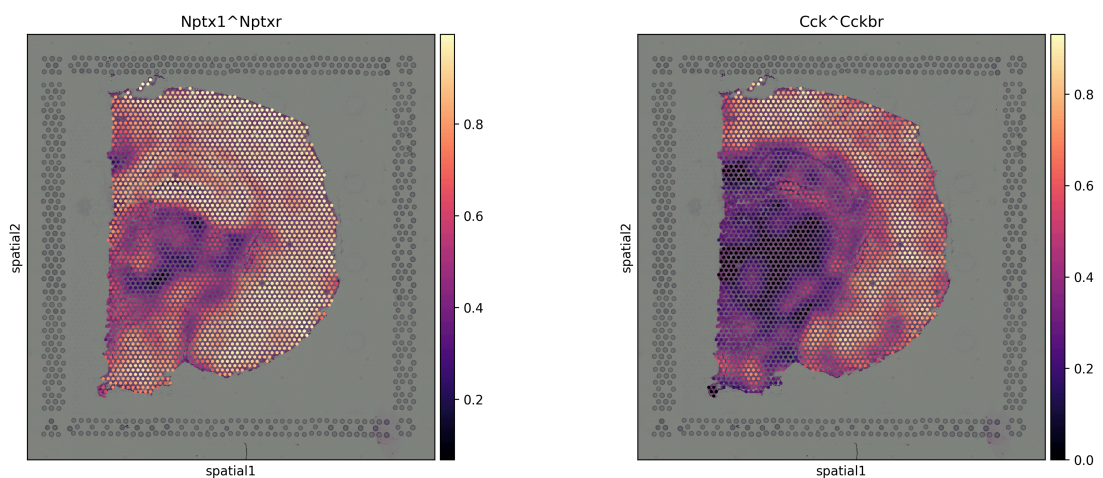


```
In [35]: # several panels -> a list of axes; all share one figure
axs = sq.pl.spatial_scatter(lrdata, color=["Nptx1^Nptxr", "Cck^Cckbr"], c
axs[0].figure.savefig(root_figure / "liana_spatial_LR.png", dpi=300, bbox_inches='tight')
plt.show()
```

```
... storing 'ligand' as categorical
... storing 'receptor' as categorical
```



```
In [41]: axs[0].figure.savefig(root_figure / "features_vs_gene_clusters.png", dpi=
```



Plot: two things to watch for:

1. Save before `plt.show()`. In Jupyter, `show()` closes the figure, so saving afterwards writes a blank image.
2. `bbox_inches="tight"` matters. Without it, legends placed outside the axes, like the ones you move below the plot in cell 25, get cut off.

```
In [ ]:
```

Save spatial LIANA object

- That way you don't have to rerun `li.mt.bivariate`.
- `adata.uns["liana_res"]` is already saved inside `visium_hne_complete.h5ad` if you write that file after running LIANA.

```
In [42]: lndata.write_h5ad(root_data / "visium_hne_liana_bivariate.h5ad")
```